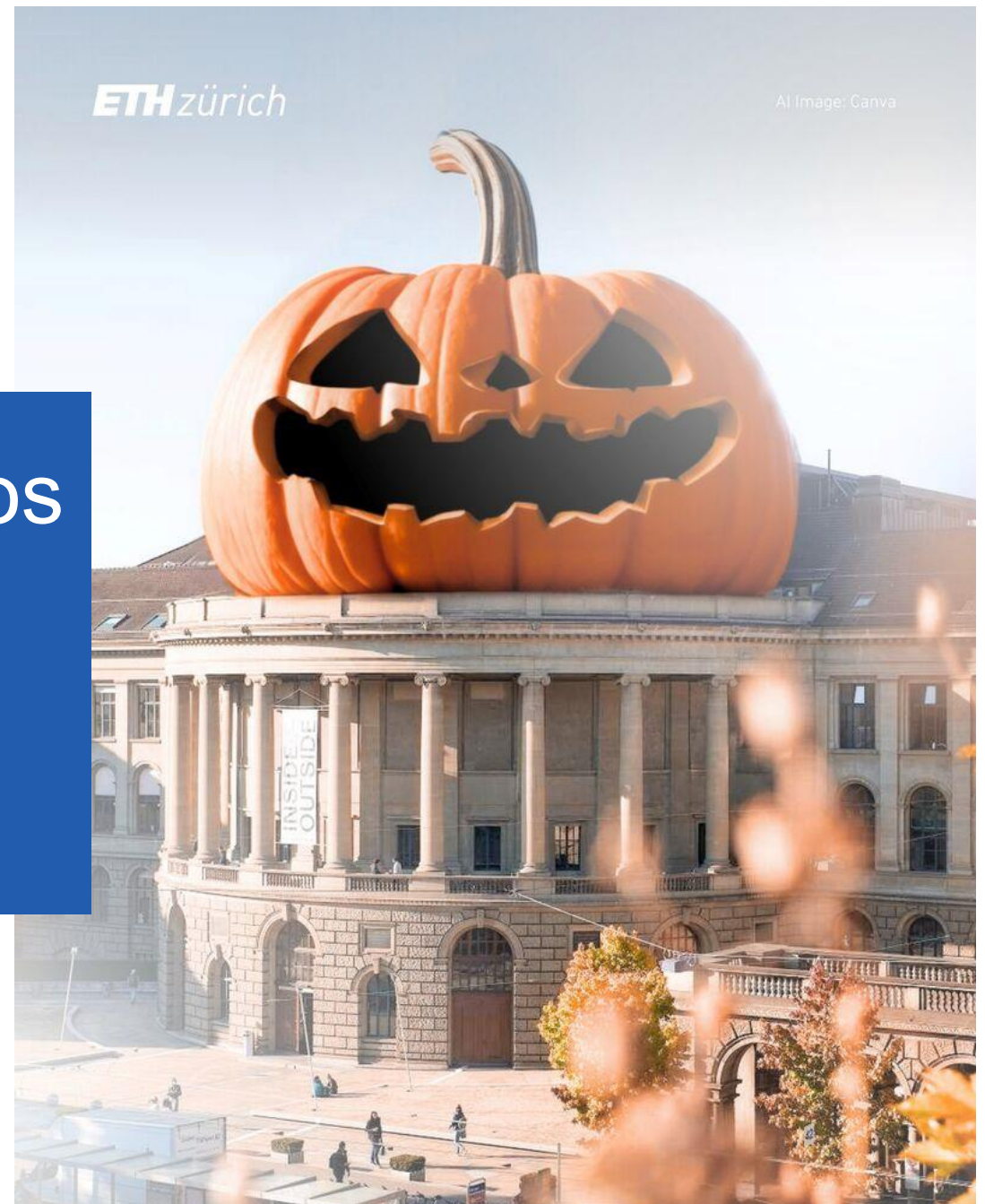


Estimating Solvency Ratios Using Machine Learning

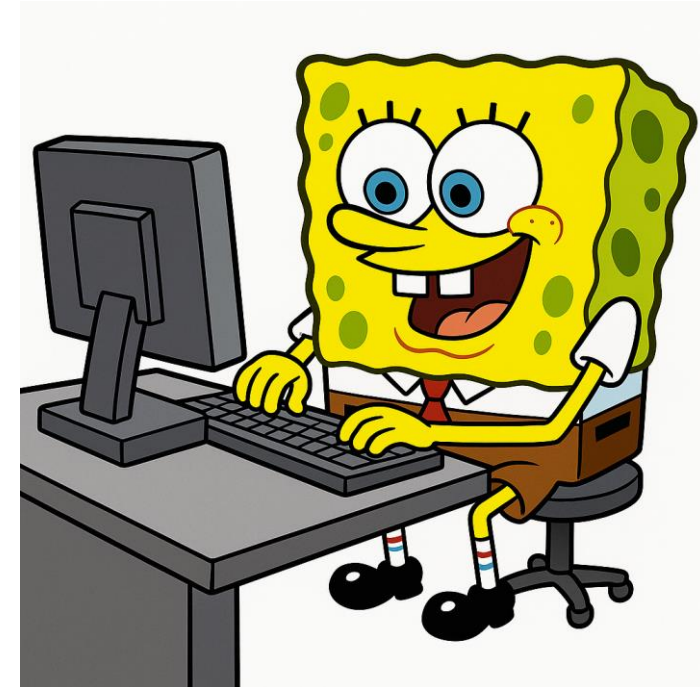
Oliver Stoll, Mahbod Tajdini, Santiago Brunner
CAS Innovation Project
07.11.2025



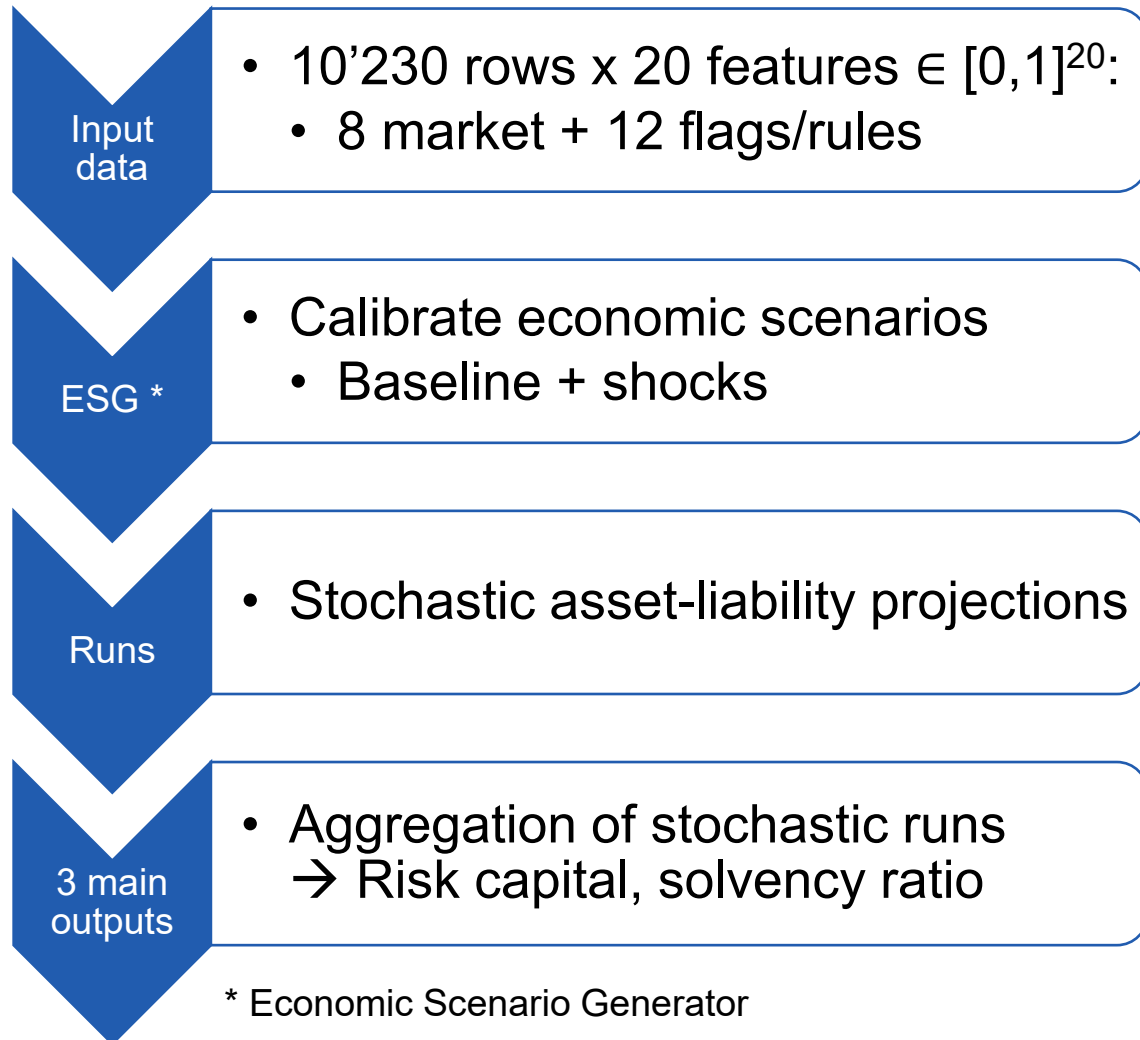
We want to forecast solvency ratios for life insurance over a short time horizon

- Solvency II / Swiss Solvency Test: reporting regime in the EU and Switzerland for insurance companies
- **Solvency ratio** = $\frac{\text{Available capital}}{\text{Required capital}} \gg 100\%$
- Risk management: **continuous** monitoring of solvency situation
- **Problem:** Full actuarial runs are slow & costly
- **Idea:** Use ML tools to learn **dependencies** between **economic market conditions** and **solvency ratio** *
- **Solution:** Feed new market parameters into ML model and instantly get a new solvency estimate

* Underlying assumption: short-term movements of ratio mainly driven by external market rates

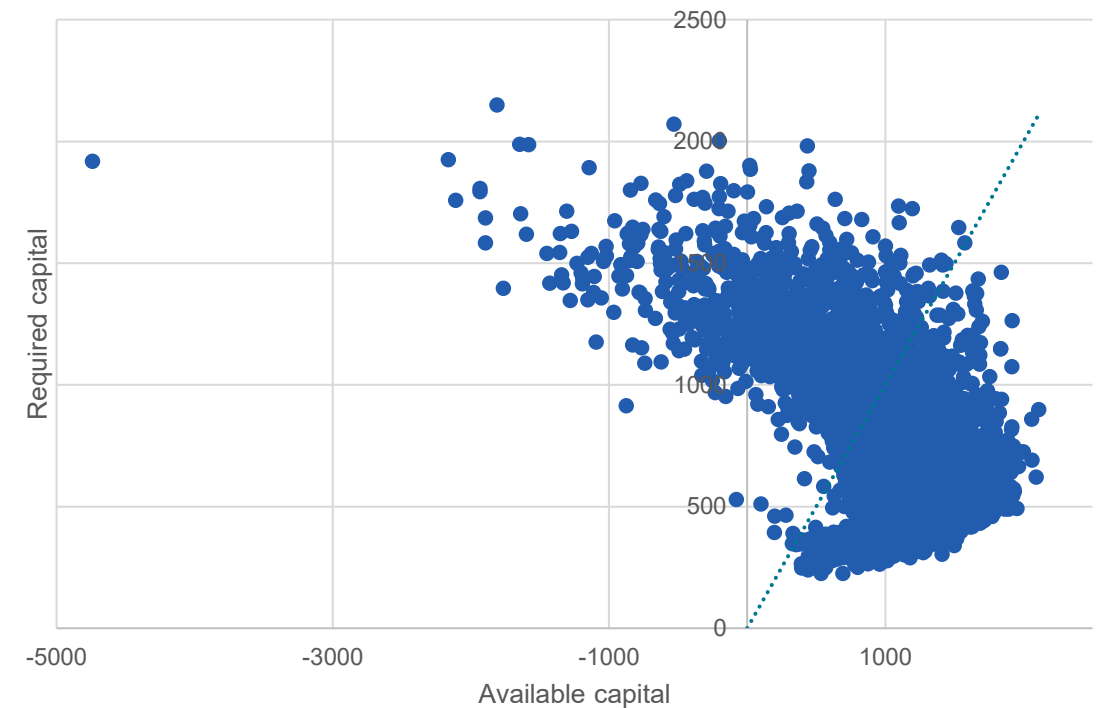


Generating the synthetic input data required a lot of effort

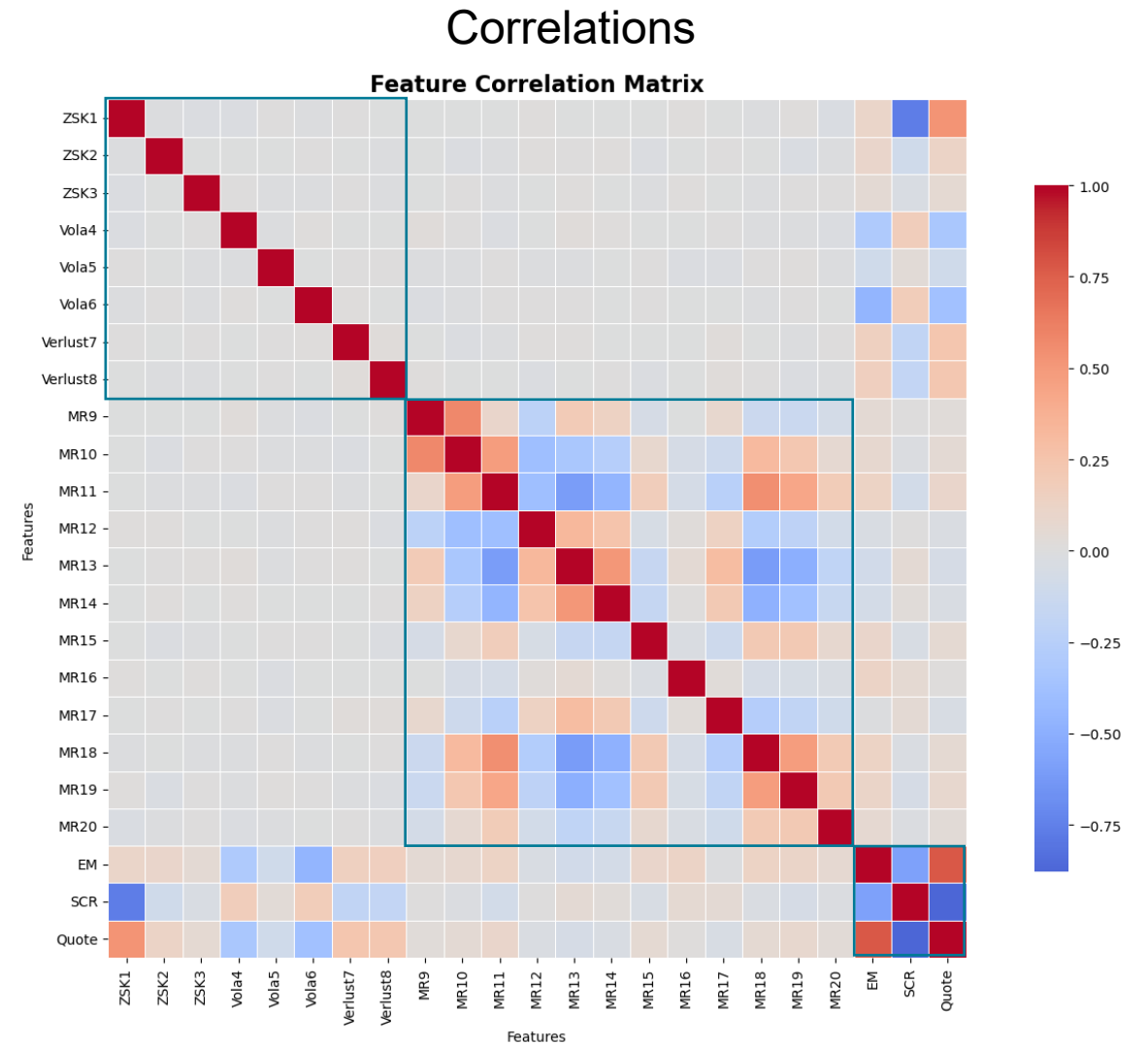
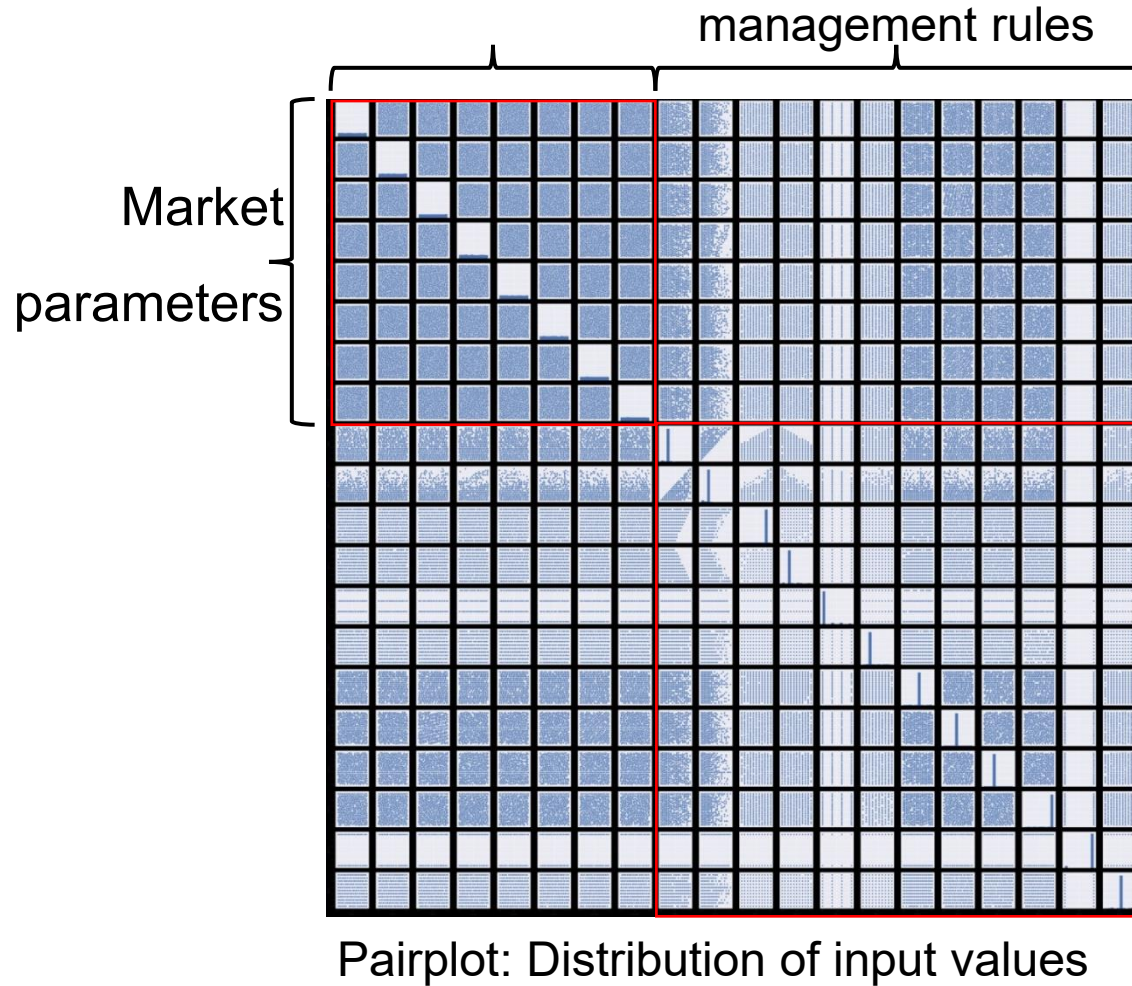


10'230 x 3 outputs

Required vs. available capital



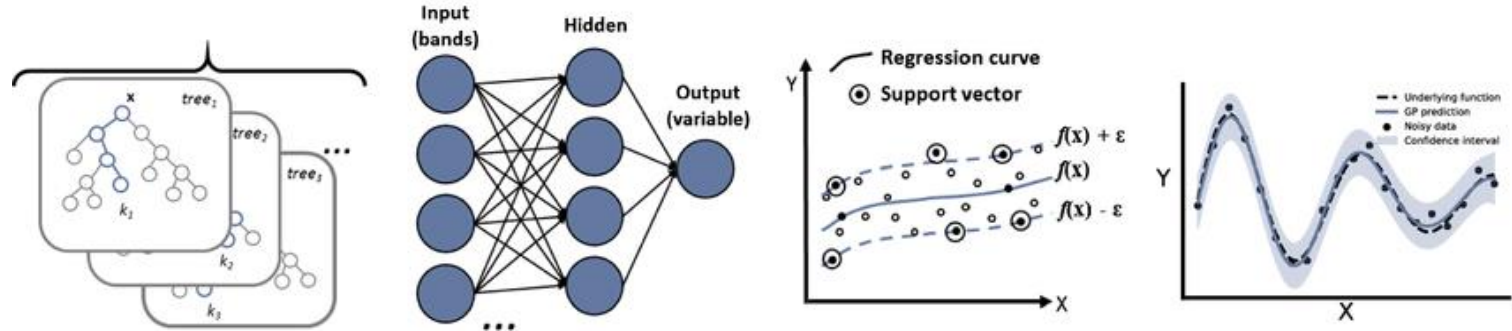
Market parameters in data set:
Covering the full $[0,1]$ intervals and no correlation.



We explored a large number of model families and setups

Model families:

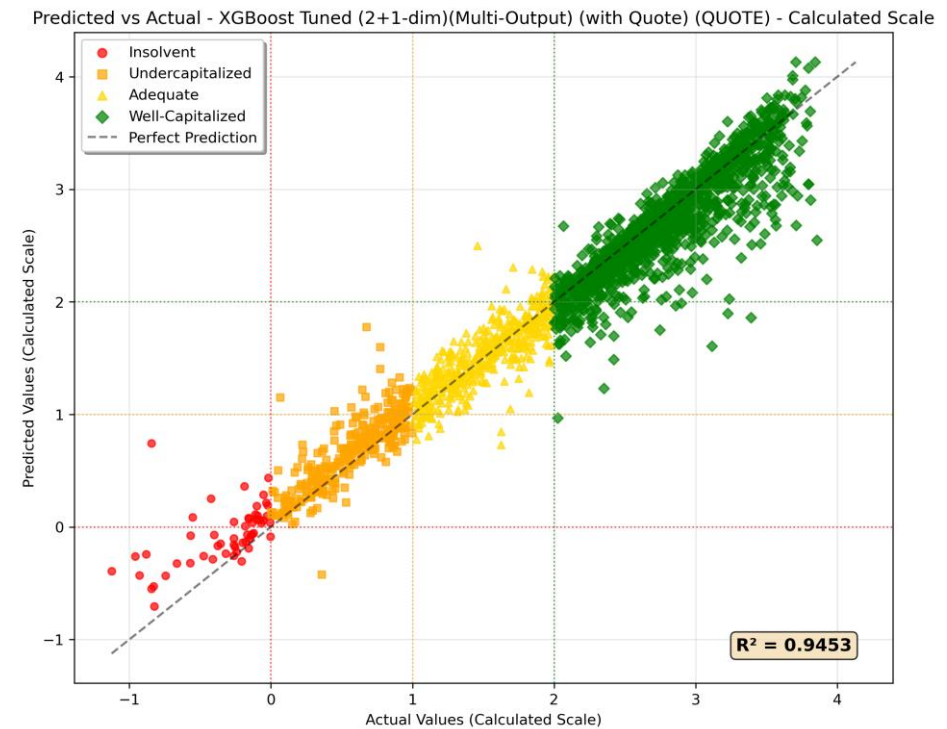
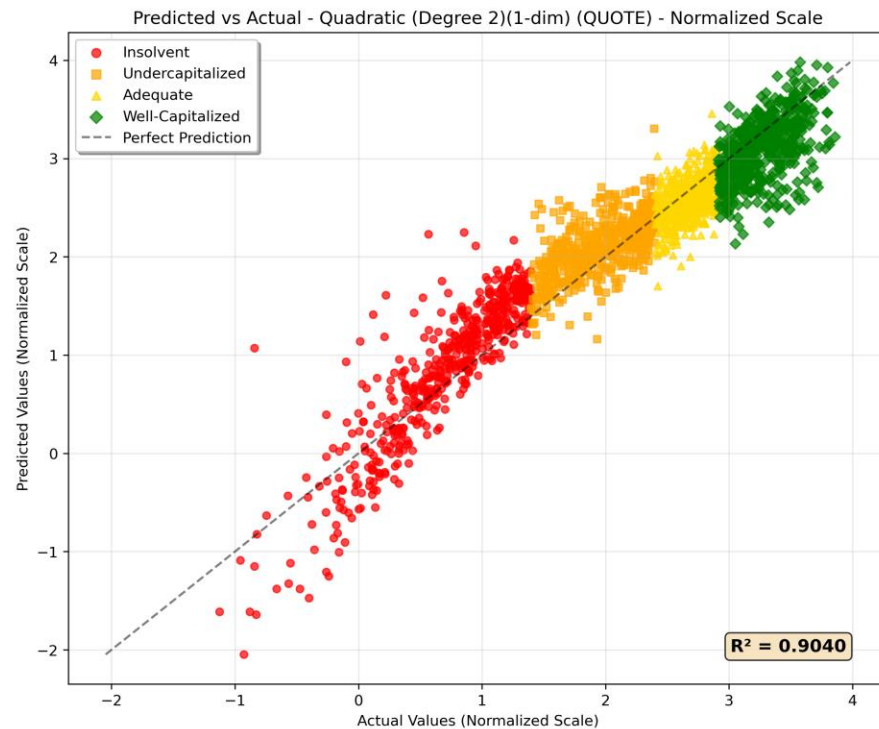
- **Regression:** various linear models plus quadratic and cubic
- **Tree-based** models: Random Forest and XGBoost
- **Neural networks**



- **Grid search** for architecture / hyperparameters
- **Direct prediction** of solvency ratio vs. **indirect prediction** (= available / required capital)
 - Regression performed better on direct prediction
 - Indirect preferred as helps understanding of drivers for solvency ratio

XGBoost and neural networks outperformed other model classes

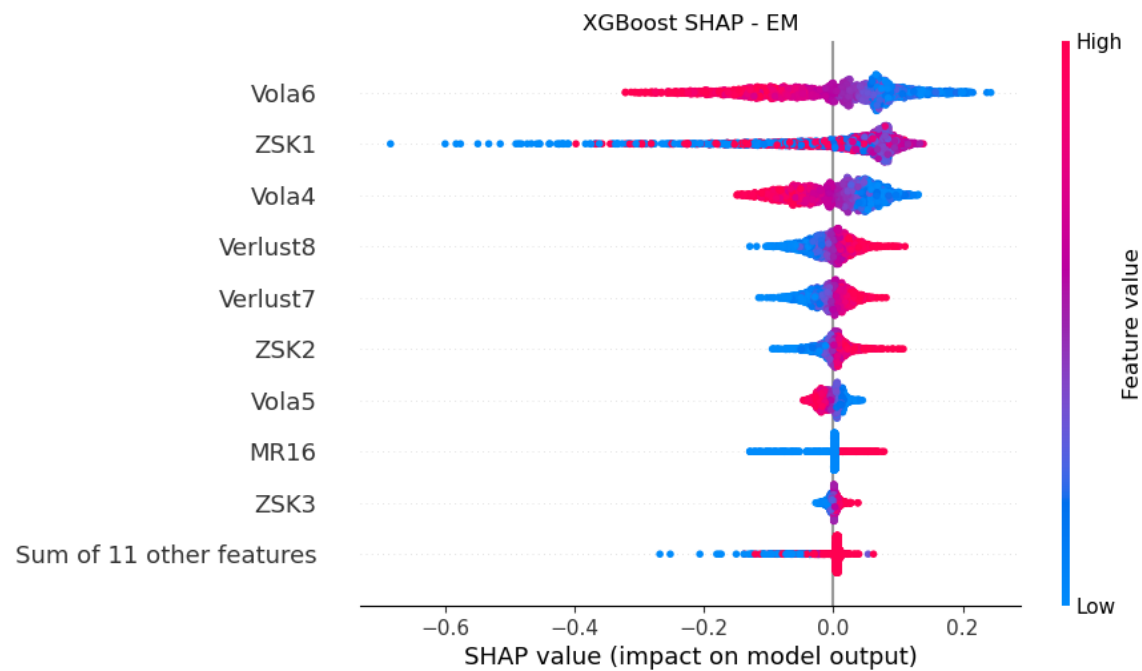
Predicted vs. actual plots for solvency ratio (perfect fit on line of identity)



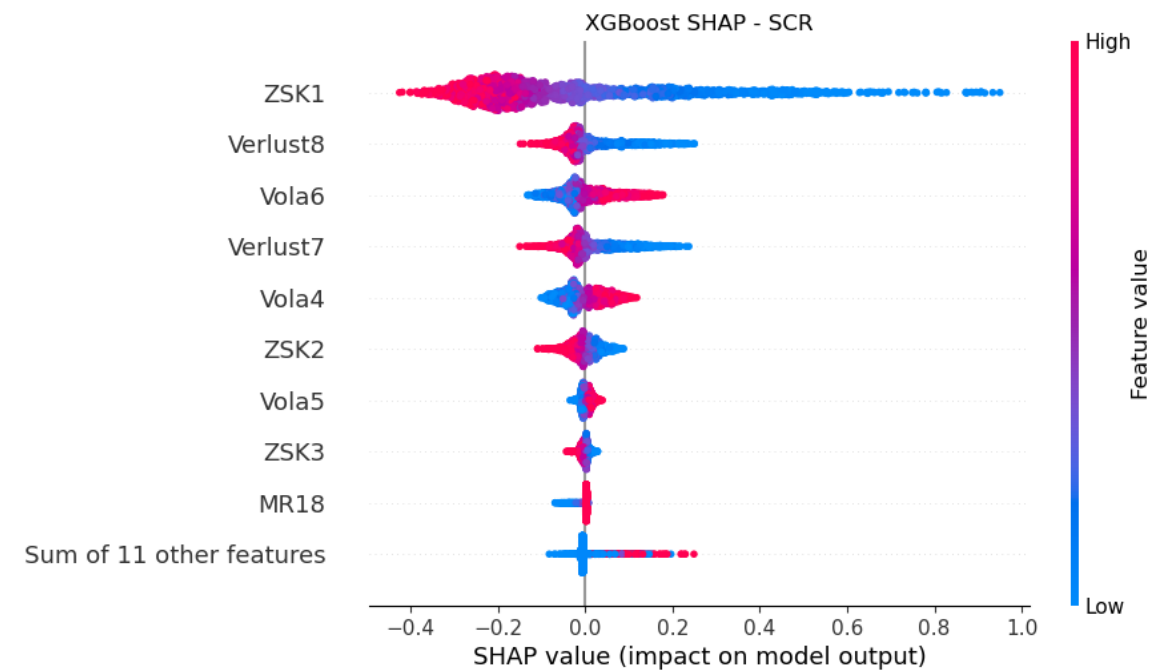
Best regression model (quadratic) vs. Gradient Boosted Trees

Explainability through SHAP values

- ZSK1 ($\approx$ **interest rate level**) by far the most important feature
→ In line with actuarial intuition

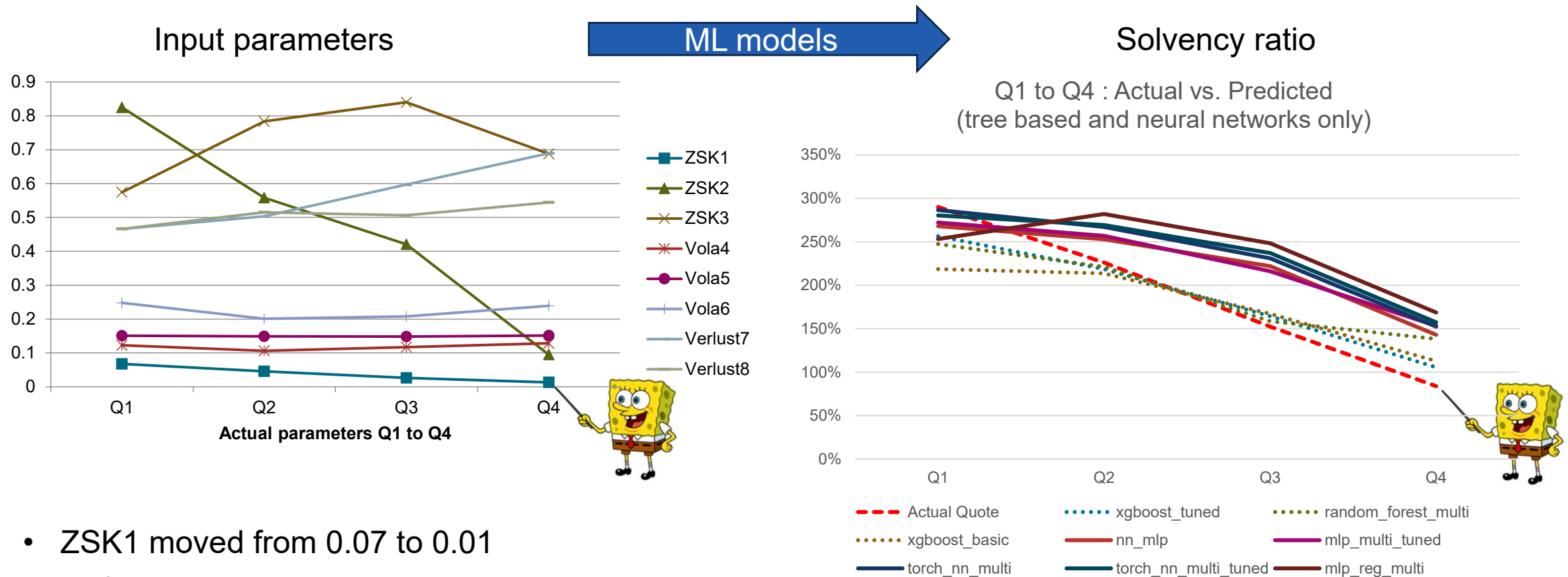


Beeswarm for available capital



Beeswarm for required capital

Back testing: XGBoost predicted observed change better than NN

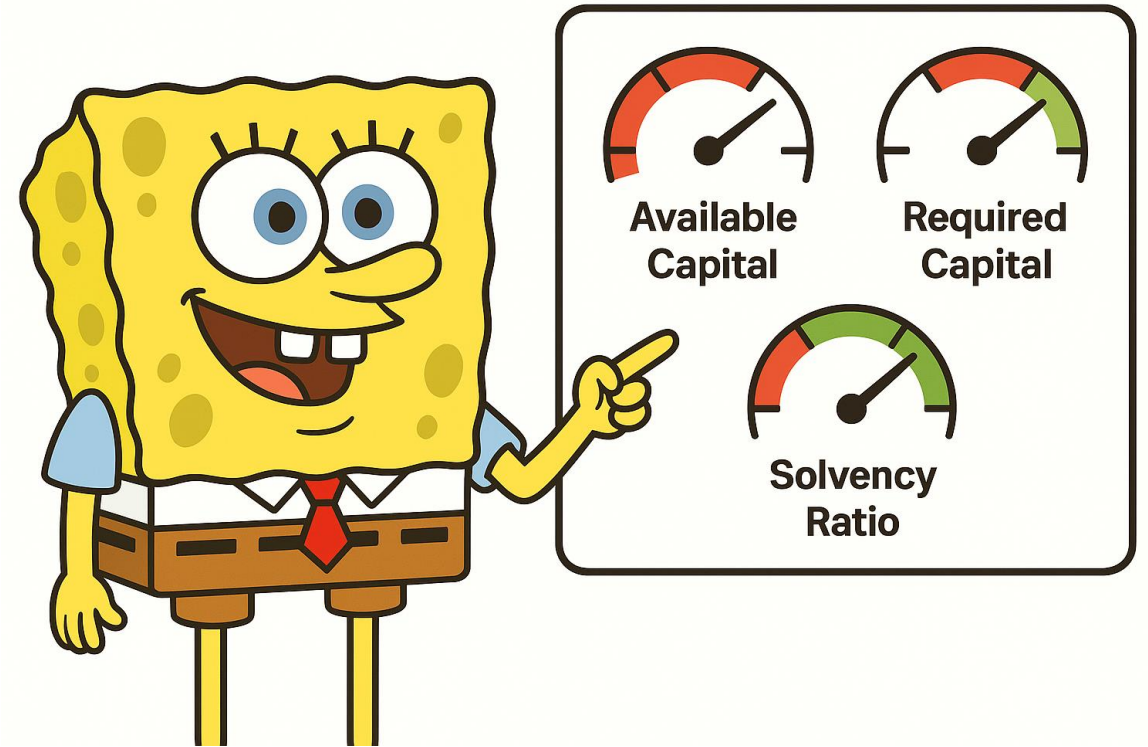


- ZSK1 moved from 0.07 to 0.01
 - SHAP: Available capital: ↓, required capital: ↑
- Ratio was dragged down from 290% to 84%

Use case: Dashboard for solvency ratio movement

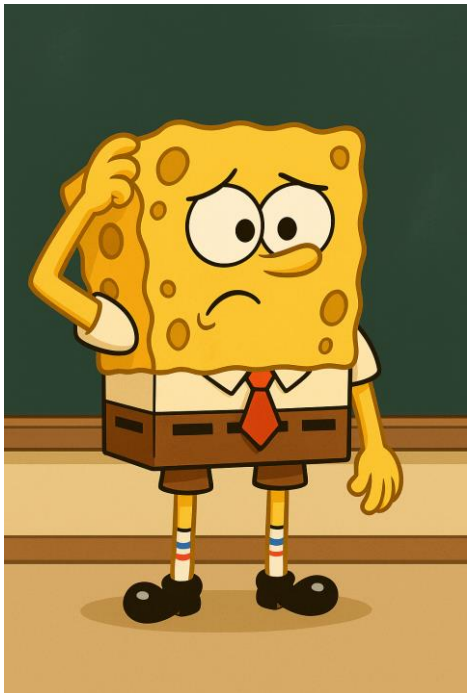
Components:

- Estimate available capital
- Estimate required capital
- Estimate solvency ratio
- Monitor:
 - Δ Economic inputs (fast-moving)
 - Δ Asset allocation (potentially fast-moving)
 - Δ Insurance business composition (slow-moving)
- Caveats:
 - Only indicator, not regulatory metric
 - Stability of model must be verified over multiple reporting periods (more research needed)



Many thanks to the two master students who supported me:

- Santiago Brunner
- Mahbod Tajdini



Time for Q&A (if there is time...)



Extra points if you can recognise to which album cover the guitar refers to...